

CPE 486/586: Deep Learning for Engineering Applications

05 Deep Learning for Sequence Data

Spring 2026

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Outline

1. Motivation
2. Convolutional Neural Network
3. Modern Convolutional Neural Network

Motivation

0	0	1	0	0	1	1	0	0	0	1	1	0	1	0	1	1	0	0	1
1	1	1	1	0	0	0	1	0	0	1	1	0	1	1	1	0	0	1	1
1	0	1	0	0	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0

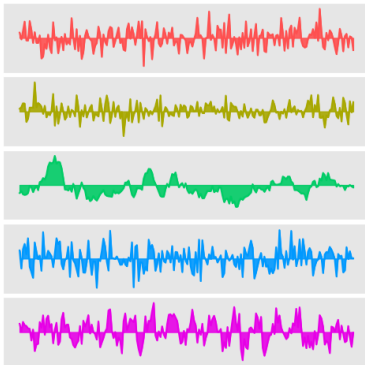
Until Now ...

- 1 Data modality: Non-sequential (tabular with no order for samples or features)
- 2 Process modality: iterative refinement of solution, but not necessarily sequential (non-generative processes – hence no notion of sequential)

Sequential Data



Image/Video



Timeseries

**Efficient modelling
of feature
interactions
underpins**

Texts/Natural
Language/Computer Code

Can we use what we learnt so far on Sequential Data?

Short answer: **May be!** But not quite!

So where is the problem?

Underlying Assumptions

- 1 Core assumptions have been that data samples are iid (independent and identically distributed)
- 2 There are also no temporal or spatial dependencies (order of features in the tabular data didn't matter until now).
- 3 Generally feature length (or number of features are fixed).
- 4 Non-stationarity assumption: data distribution didn't change.

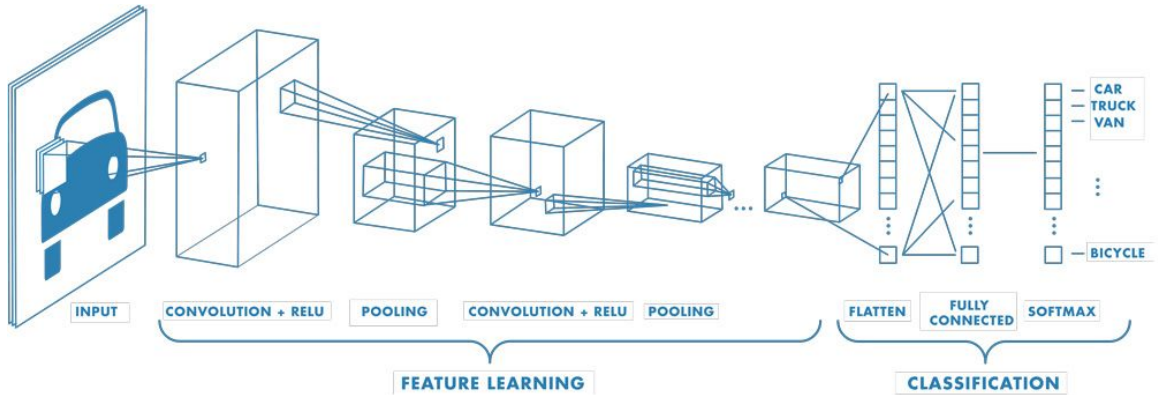
Learning on Sequential Data

- 1 Images: Convolutional Neural Network (CNN)
- 2 Texts/Natural language: Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM) Network, Gated Recurrent Unit (GRU) Network, Transformer
- 3 Videos: CNN + LSTM/GRU/Transformer

Convolutional Neural Network

0	0	0	0	1	1	0	0	0	0	0	0	1	0	0	1	1	0	1	0
1	1	1	0	0	0	1	1	0	0	1	0	0	1	0	0	1	0	0	1
1	0	1	0	1	1	1	1	1	0	1	1	0	1	0	1	1	1	1	1

Convolutional Neural Network

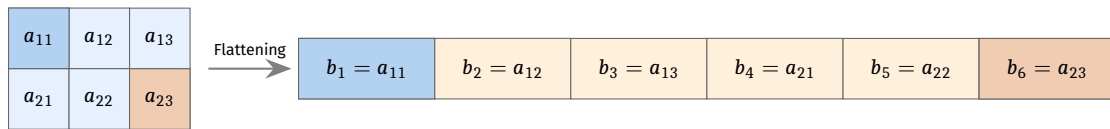


Source:

<https://www.mathworks.com/discovery/convolutional-neural-network.html>

Motivation for CNN

⚡ **(Fully-connected Neural Network, FNN Limitation:** Spatial data must be converted to a **1-D array** (Flattening).



- ✗ **Spatial Loss:** We lose the **pixel-to-pixel relationship** and local geometry.
- ✗ **RGB Complexity:** In addition, working with color images would mean we have at least three channels, R, G, B.
- ✗ **Parameter Explosion:**
 - $200 \times 200 \times 3 = 120,000$ features per image.
 - Just 100 images = **12,000,000** features to process!

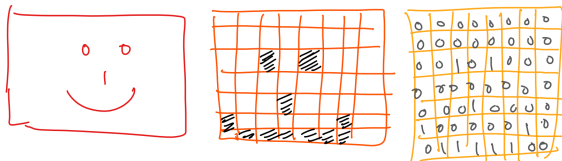
Limitations of FNN for Images

In that sense, FNN suffers from two conditions:

- ⚡ Unable to preserve spatial information
- ⚡ Curse of dimensionality

To preserve the pixel-to-pixel spatial relationship, we include two additional operations in the neural network:

- 1 Convolution
- 2 Pooling



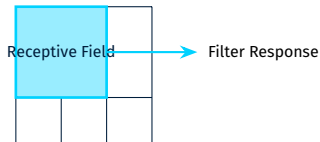
What is a CNN?

Beyond Flattening

A **Convolutional Neural Network (CNN)** is designed for grid-structured data. Unlike FNNs, it treats images as spatial volumes, preserving local relationships.

The Architectural Edge:

- ⚡ **Local Receptive Fields:** Neurons only “see” a small local region, mimicking the human visual cortex.
- ⚡ **Spatial Hierarchy:** Layers learn to recognize edges, then shapes, then objects.



Weight Sharing & Pooling

Weight Sharing

Filters share weights across the entire image.

- ⚡ **Efficiency:** Drastically reduces parameters.
- ⚡ **Feature Detection:** A vertical edge detector works everywhere!

Pooling

Subsampling to reduce spatial dimensions.

- ⚡ **Invariance:** Stable against small translations/shifts.
- ⚡ **Speed:** Lowers computational load & memory.

FNN: 224x224 image $\approx$ 150k+ weights per neuron.

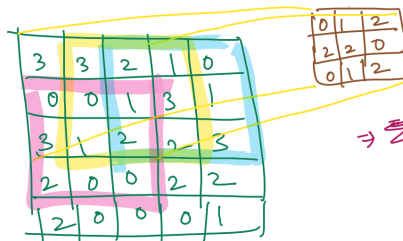
CNN: 3x3 filter = **9 weights** per feature map.

Building blocks: Convolution

Convolution is an operation to reduce the number of pixels through some operation that, in turn, reduces the number of weights to learn.

In convolution, the reduction in the number of features happens by sliding a kernel across an image, performing the dot product.

Kernel: It is an $n \times n$ matrix of some numbers whose dimension is usually smaller than the image size.



5x5 matrix

$$\sum \begin{bmatrix} 3 \times 0 & 3 \times 1 & 2 \times 2 \\ 0 \times 2 & 0 \times 2 & 1 \times 0 \\ 2 \times 0 & 1 \times 1 & 2 \times 2 \end{bmatrix}$$

$$\Rightarrow \sum [0 + 3 + 4 + 0 + 0 + 0 + 0 + 1 + 4] = 12$$

$$\begin{bmatrix} 12 & 12 & 17 \\ 10 & 17 & 19 \\ 9 & 6 & 14 \end{bmatrix}^{3 \times 3}$$

Convolution Using Kernel

3	3	2	1	0
0	0	1	3	1
3	1	2	2	3
2	0	0	2	2
2	0	0	0	1

5x5 Input Matrix

3x3 Kernel

0	1	2
2	2	0
0	1	2

$$\begin{aligned} \sum & \begin{bmatrix} 3 \times 0 & 3 \times 1 & 2 \times 2 \\ 0 \times 2 & 0 \times 2 & 1 \times 0 \\ 3 \times 0 & 1 \times 1 & 2 \times 2 \end{bmatrix} \\ &= 0+3+4+0+0+0+0+1+4 \\ &= 12 \end{aligned}$$

12	12	17
10	17	19
9	6	14

3x3 Output Map

Kernels

A particular kernel usually extracts certain features from an image, or basically, it performs some sort of image processing.

Examples of Kernels

$$\text{Edges: } \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\text{Sharpen: } \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\text{Box Blur: } \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Edge-detection Example

Original Image



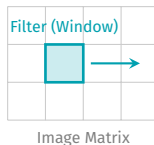
Edge Detection Filtered Image



Filter

The window we use in kernel is called a **filter**. The window of a certain size is slid across the image. We can apply pooling to perform operations such as **minimum**, **median**, **maximum**.

- ⚡ **Images** can be represented as **matrices**, where each element corresponds to a pixel.
- ⚡ A **filter** is just a small matrix that is **convolved** with same-sized sections of the image matrix.



Convolutional Filters

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \dots & \dots \\ \vdots & \ddots & \\ \vdots & & \ddots \end{bmatrix}$$

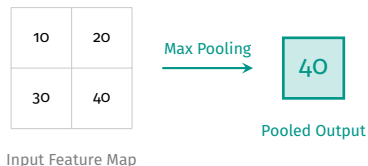
$$\begin{aligned} & (0 \times 0) + (0 \times 1) + (0 \times 0) + \\ & (0 \times 1) + (1 \times -4) + (2 \times 1) + \\ & (0 \times 0) + (2 \times 1) + (4 \times 0) = \mathbf{0} \end{aligned}$$

Convolutional Filters

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -1 & 0 \\ -2 & -5 & -5 & -2 \\ 2 & -2 & -1 & 3 \\ -1 & 0 & -5 & 0 \end{bmatrix}$$

Pooling

Pooling does a mathematical operation on the output of the convolution image. But you can also apply pooling directly to the image.



Building Blocks: Pooling

Pooling is a mathematical operation on the output of the convolution image. But you can also apply pooling directly to the image.

Pooling: Downsampling

Combine multiple adjacent nodes into a single node: max-pooling

$$\begin{bmatrix} 0 & -1 & -1 & 0 \\ -2 & -5 & -5 & -2 \\ 2 & -2 & -1 & 3 \\ -1 & 0 & -5 & 0 \end{bmatrix} \rightarrow \text{max} \rightarrow \begin{bmatrix} 0 \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} 0 & 1 & -1 & -0 \\ -2 & -5 & -5 & -2 \\ 2 & -2 & -1 & 3 \\ -1 & 0 & -5 & 0 \end{bmatrix} \rightarrow \text{max} \rightarrow \begin{bmatrix} 0 & 0 \end{bmatrix} \quad (2)$$

- ⚡ Reduces the dimensionality of the input to subsequent layers and thus, the number of weights to be learned
- ⚡ Further, it protects the network from (slightly) noisy inputs

Downsampling with Stride

⚡ Only apply the convolution to some subset of the image

⚡ e.g., every other column and row = a **stride of 2**

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} -2 & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

Downsampling with Stride

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} -2 & -2 & . \\ . & . & . \\ . & . & . \end{bmatrix}$$

Downsampling with Stride

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} -2 & -2 & 1 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

Downsampling with Stride

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} -2 & -2 & 1 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

Operation after Pooling

- ⚡ After applying convolution and pooling, we apply a non-linear activation, **ReLU**.
- ⚡ Then after that, you have more Convolution, more Pooling layers.
- ⚡ Finally, after several sequences of **Conv + Pool + ReLU**, we **flatten** the output and feed it to the fully connected layer (ANN).

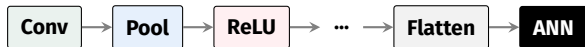
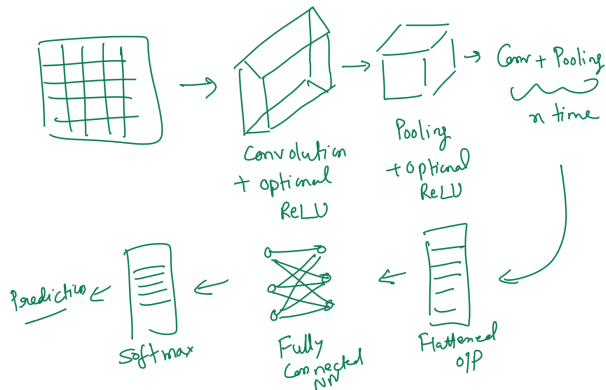
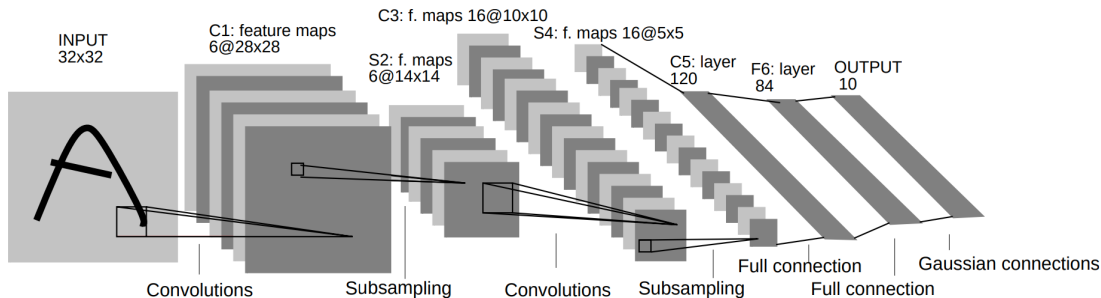


Image Classification

Then, we apply a softmax activation if our goal is to do supervised image classification.



Lenet

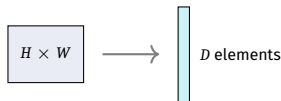


Source: http://vision.stanford.edu/cs598_spring07/papers/Lecun98.pdf

- ⚡ One of the earliest, most famous deep learning models – achieved remarkable performance at handwritten digit recognition ($< 1\%$ test error rate on MNIST dataset)
- ⚡ Used sigmoid (or logistic) activation functions between layers and mean-pooling, both of which are pretty uncommon in modern architectures

A note on multi-dimensionality and vectorization

- ⚡ **High-Dimensionality:** Images are high-dimensional datasets. This is true even more for **colored images**.
- ⚡ Let $X \in \mathbb{R}^{H \times W}$ be a matrix (for a grayscale image of height H and width W).
- ⚡ **Flattening:** After flattening, an image is represented as a column vector $X \in \mathbb{R}^D$:
 - **Grayscale:** $D = H \times W$
 - **Color (3 channels):** $D = H \times W \times 3$



Tensor

We need a concept of higher-order matrices called a **Tensor**.

⚡ $X \in \mathbb{R}^{H \times W \times D}$ is an **order-3 tensor**. Consider it as D channels of matrices. (Note: $D = 1$ reduces it to a standard matrix).

Tensor Orders

- ⚡ **Order-0:** A Scalar value.
- ⚡ **Order-1:** A Vector.
- ⚡ **Order-2:** A Matrix. (*Corrected from your text*)
- ⚡ **Order-3:** A Cube (e.g., RGB Image).

We can also have **4 dimensions** if we consider opacity (e.g., RGBA in PNG).

Tensor to Vector

Given a tensor, we can arrange all numbers into a long vector following a **prespecified order**.

Example (Matrix Vectorization - Column First Order):

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{\text{MATLAB } A(:)} \begin{bmatrix} 1 \\ 3 \\ 2 \\ 4 \end{bmatrix}$$

Recursive Vectorization

To vectorize an **order-3 tensor**, we vectorize the first channel (matrix), then the second, and so on. This recursive process applies to tensors of any order > 3 .

Tensor in CNN

In CNNs, the convolution kernel will form an **order-4 tensor**.

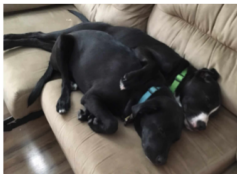
CNN Input Dimensions

- 1 **Depth:** Number of channels in a colored image.
- 2 **Filters:** Multiple filters learning specific features.
- 3 **Spatial Dimensions:** $H \times W$ of the image.

Tensor Orders (Examples):

- ⚡ **Order 3:** A **batch** of grayscale images.
- ⚡ **Order 4:** A **batch** of colored images or a single colored movie.
- ⚡ **Order 5:** A **batch** of movies.

Tensors



Example:
 $3 \times 4 \times 6$ tensor

4	1	2	16	3	6						
1		5	2	6	14	15	8				
5		26									
0		0		4	6	8	9	5	3		
				16	5	2	8	2	1		
		7		5	2	14	11	7	8		
				15	2	5	0	9	8		

Source: <https://www.cs.cmu.edu/~mgormley/courses/10601//slides/lecture17-cnn.pdf>

Further Reading

<https://www.cs.toronto.edu/~hinton/absps/NatureDeepReview.pdf>

http://vision.stanford.edu/cs598_spring07/papers/Lecun98.pdf

Modern Convolutional Neural Network

1	1	1	0	1	0	1	1	0	1	0	1	1	1	0	0	0	1	0	
1	0	0	1	1	1	0	1	1	1	1	0	0	1	0	0	0	1	1	0
0	0	1	1	1	1	0	0	1	0	1	1	0	0	1	1	0	1	0	1

Modern Days of CNN Had Arduous Path

- 1 CNN didn't find its popularity easily.
- 2 There was a **lack of good datasets**.
- 3 **Kernel and ensemble methods** were still better until 2012.
- 4 Computer vision pipelines used methods such as *SIFT*, *SURF*, *bag of visual words*.
- 5 Researchers were using **hand-crafted features**.

The Paradigm Shift

However, **LeCun, Hinton, Ng**, and others thought features can be learnt too. **AlexNet** (after Alex Krizhevsky) proved that.

Finally Some Interesting and Useful Datasets

⚡ Large-Scale Benchmarks

- 1 **ImageNet (2009):** <https://huggingface.co/datasets/ILSVRC/imagenet-1k>
- 2 **CIFAR-10 & CIFAR-100:** <https://www.cs.toronto.edu/~kriz/cifar.html>
- 3 **LAION Dataset:** <https://laion.ai/projects/>

In addition, Yan LeCun used **MNIST**:

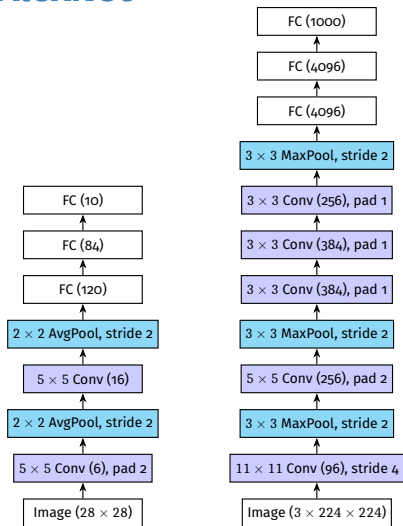
https://github.com/rahulbhadani/CPE490_590_Sp2025/tree/master/Data/MNIST

We also have **FashionMNIST**:

<https://github.com/zalandoresearch/fashion-mnist>

Note: These two are smaller size datasets.

AlexNet



From LeNet (left), AlexNet (right).

AlexNet Architecture

- 1 Convolution window shape: 11×11 , then 5×5 , and then 3×3 .
- 2 Network adds maxpooling of 3×3 with a stride of 2.
- 3 At the end, two huge fully connected layers with 4096 outputs.
- 4 ReLU activation function.

Block Thinking

Neurons



Layers



Blocks

Basic building blocks of CNN are sequence of **(i)** Convolutional layer with padding **(ii)** a nonlinearity such as a ReLU **(iii)** a pooling layer such as max-pooling to reduce the resolution.

But pooling layer reduces dimension by $\log_2 d$ for d dimension input, and it can happen fast exponentially.

- 1 **Simonyan and Zisserman (2014)** used multiple convolutional layer in between downsampling via max-pooling in the form of block.
- 2 **Two 3×3 convolution** have same effect as one 5×5 convolution layer: **deep and narrow networks significantly outperform their shallow counterparts.**

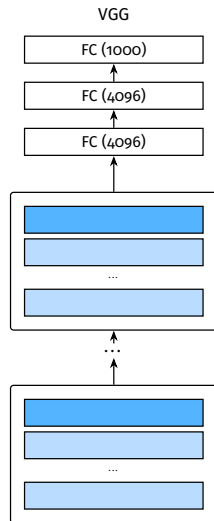
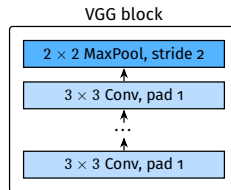
Three important papers in that area:

- 1 Simonyan, K. & Zisserman, A. "Very deep convolutional networks for large-scale image recognition." (2014).
- 2 Lavin, A. & Gray, S. "Fast algorithms for convolutional neural networks." (2016).
- 3 Liu, Z. et al. "A convnet for the 2020s." (2022).

VGG Block

VGG Block (**Simonyan and Zisserman (2014)**), consists of a sequence of convolutions with 3×3 kernels with padding of 1 (keeping height and width) followed by a 2×2 max-pooling layer with stride of 2 (halving height and width after each block).

- ⚡ VGG Network can be partitioned into two parts: the first consisting mostly of convolutional and pooling layers and the second consisting of fully connected layers that are identical to those in AlexNet.
- ⚡ But the convolutional layers are grouped in nonlinear transformations that leave the dimensionality unchanged, followed by a resolution-reduction step.



Network in Network (NiN)

Two main problems with AlexNet and VGGNet:

- 1 FCN at the end of the network has tremendous number of parameters.
- 2 Cannot add FCN earlier in the network otherwise it can destroy spatial information.

NiN:

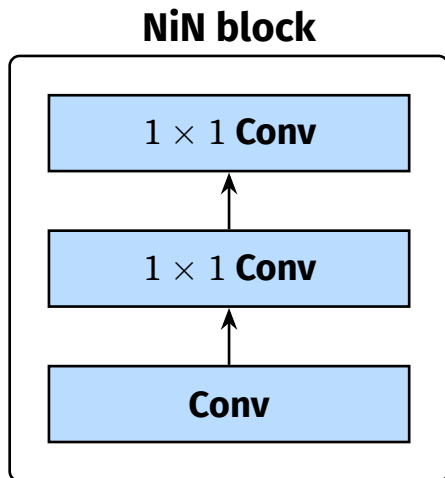
- 1 use 1×1 convolutions to add local nonlinearities across the channel activations
- 2 use global average pooling to integrate across all locations in the last representation layer

NiN Block

Input/Output of Convolutional layer is 4-d Tensor with no. of samples $\times$ channel $\times$ height $\times$ width.

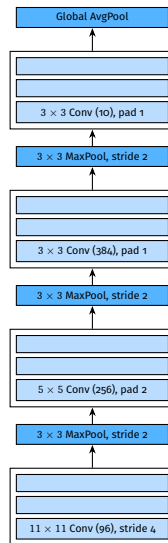
Input/output of FC layer are 2-D Tensor: no. of samples $\times$ features.

The idea behind NiN is to apply a fully connected layer at each pixel location (for each height and width). The resulting 1×1 convolution can be thought of as a fully connected layer acting independently on each pixel location.



NiN Model

- ⚡ **Initial Convolution Sizes:** NiN uses the same initial convolution sizes as AlexNet:
 - Kernel sizes are 11×11 , 5×5 and 3×3 respectively.
 - The numbers of output channels match those of AlexNet.
- ⚡ **Downsampling:** Each NiN block is followed by a max-pooling layer with a stride of 2 and a window shape of 3×3 .
- ⚡ **Architectural Shift:**
 - NiN avoids fully connected layers altogether.
 - Instead, NiN uses a NiN block with a number of output channels equal to the number of label classes.
 - Followed by a global average pooling layer, yielding a vector of logits.



GoogLeNet: Multi-Branch Networks

Szegedy, C. et al. (2015). *Going deeper with convolutions*. *Proceedings of the CVPR* (pp. 1–9).

Network Structure:

⚡ Stem for data ingestion:

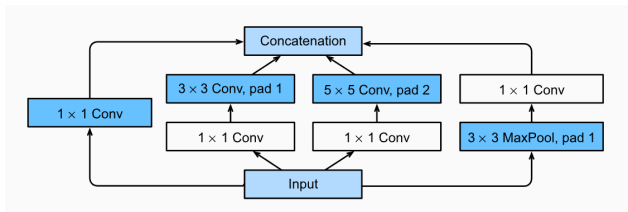
- First two or three convolutions.
- Operate directly on the image.

⚡ Body for data processing:

- Contains convolution blocks.
- It has concatenated multi-branch convolutions.

⚡ Head for prediction:

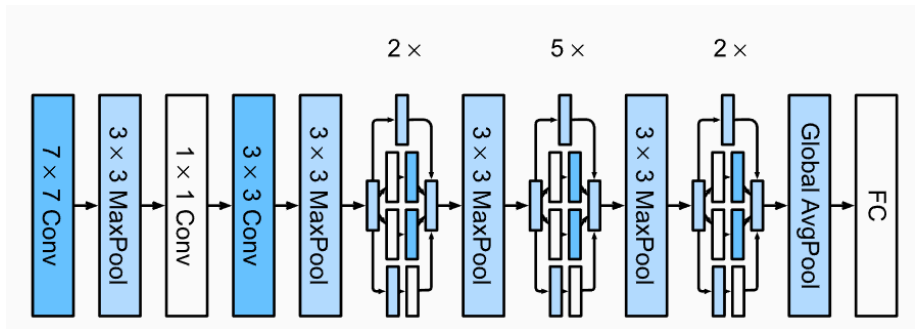
- Maps features to the required problem.
- Handles classification, segmentation, detection, or tracking.



Convolution block in GoogLeNet is the **Inception Block**.

GoogLeNet Model

GoogLeNet uses a stack of a total of 9 inception blocks, arranged into three groups with max-pooling in between, and global average pooling in its head to generate its estimates. Max-pooling between inception blocks reduces the dimensionality. At its stem, the first module is similar to AlexNet and LeNet.



Applying Batch Normalization in CNN

- ⚡ **Placement:** With convolutional layers, we can apply batch normalization after the convolution but before the nonlinear activation function.
- ⚡ **Per-Channel Logic:** The key difference from batch normalization in fully connected layers is that we apply the operation on a per-channel basis across all locations.
- ⚡ **Dimensions:**
 - Assume minibatches contain m training samples.
 - For each channel, output height is p and width is q .
 - Normalization is carried out over the $m \cdot p \cdot q$ elements per output channel simultaneously.
- ⚡ **Spatial Normalization:**
 - Collect values over all spatial locations to compute mean and variance.
 - Apply the same mean/variance within a given channel to every spatial location.
- ⚡ **Parameters:** Each channel has its own scale and shift parameters, both of which are scalars.